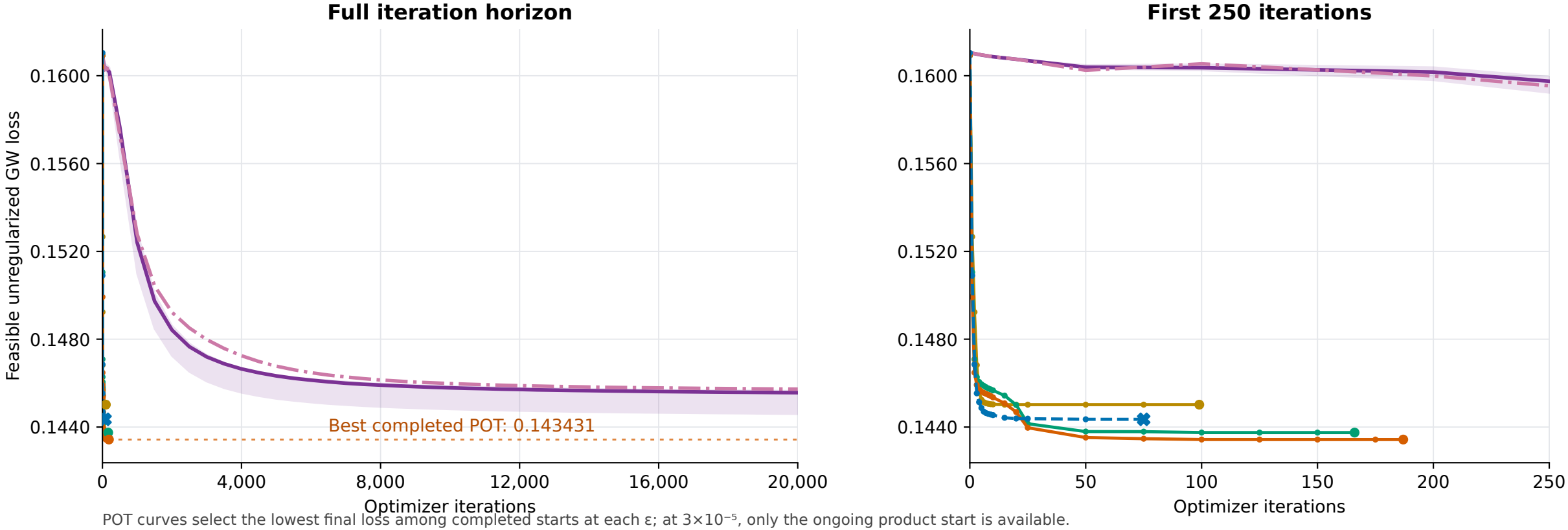
BMMC · 6,224 cells: FRAMES and the POT entropy sweep

FP64 · Feasible unregularized GW loss · Lower is better · Saved results as of 21 Oct, 09:58 Paris

FRAMES, 1,024 pairs: median of 3 seeds (band: range)
FRAMES, 4,096 pairs: CPU, seed 811
POT, ε=10⁻³: permutation mixture, seed 903
POT, ε=3×10⁻⁴: permutation mixture, seed 903
POT, ε=10⁻⁴: permutation mixture, seed 903
POT, ε=3×10⁻⁵: product start (unfinished)



POT curves select the lowest final loss among completed starts at each ε; at 3×10⁻⁵, only the ongoing product start is available.

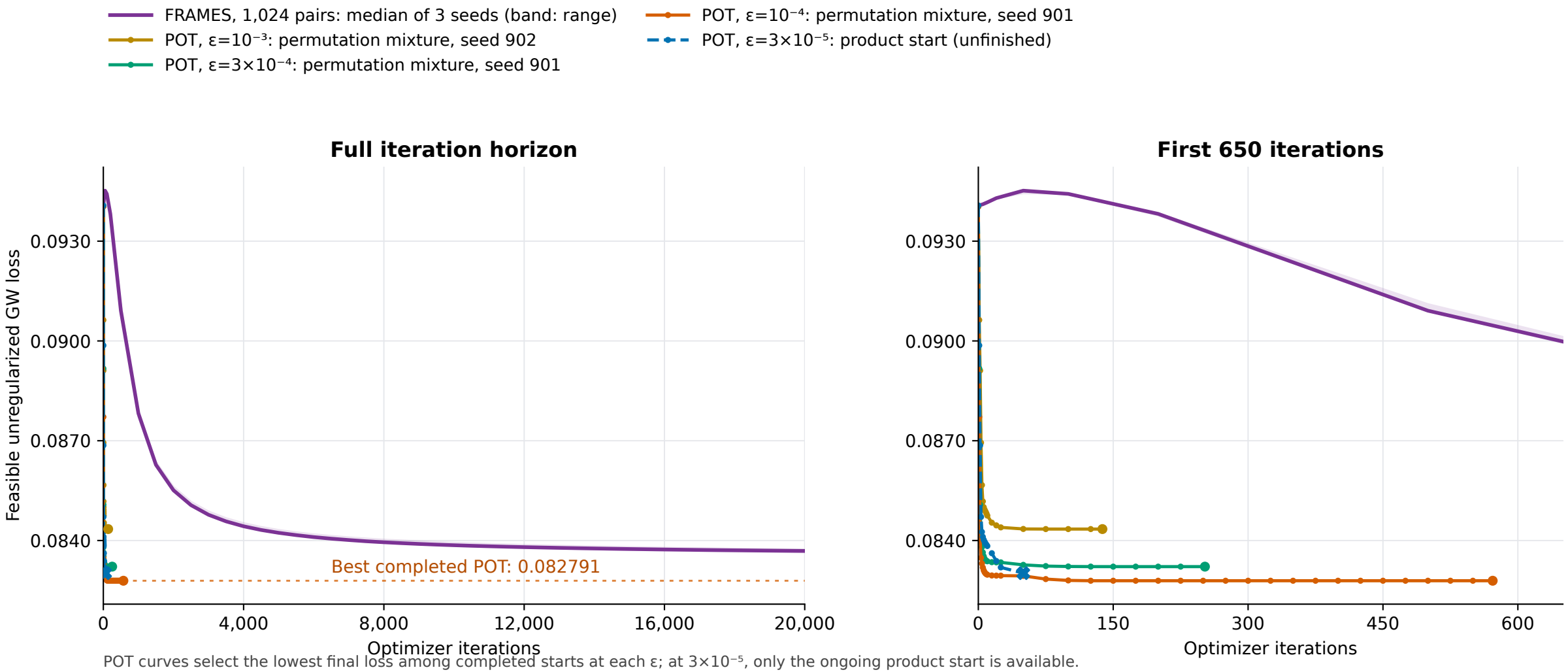
POT dots are measured losses; × marks an unfinished trajectory. The horizontal dotted line is a final-value reference only.

FRAMES uses product initialization; POT initialization varies. Iterations have different computational costs.

The 4,096-pair CPU run is one replicate; CPU/CUDA random streams differ, so batch size is not isolated.

SNARE-seq · 10,000 cells: FRAMES and the POT entropy sweep

FP64 · Feasible unregularized GW loss · Lower is better · Saved results as of 21 Oct, 09:58 Paris



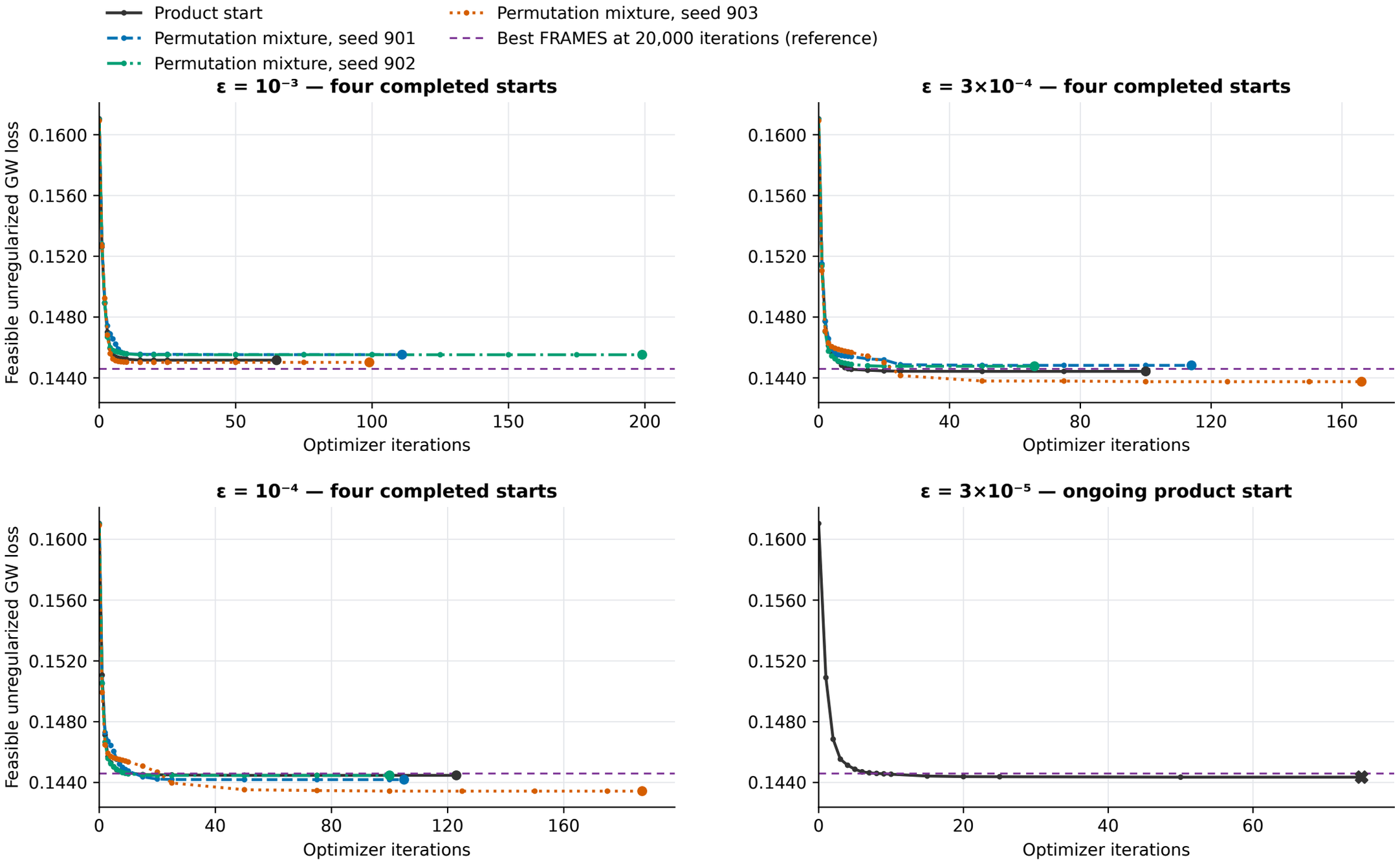
POT curves select the lowest final loss among completed starts at each ε; at 3×10^{-5}, only the ongoing product start is available.

POT dots are measured losses; × marks an unfinished trajectory. The horizontal dotted line is a final-value reference only.

FRAMES uses product initialization; POT initialization varies. Iterations have different computational costs.

BMMC · 6,224 cells: every available POT trajectory

Full-gradient entropy-regularized GW updates · Sinkhorn tolerance 10^{-7} · 21 Oct, 09:58 Paris

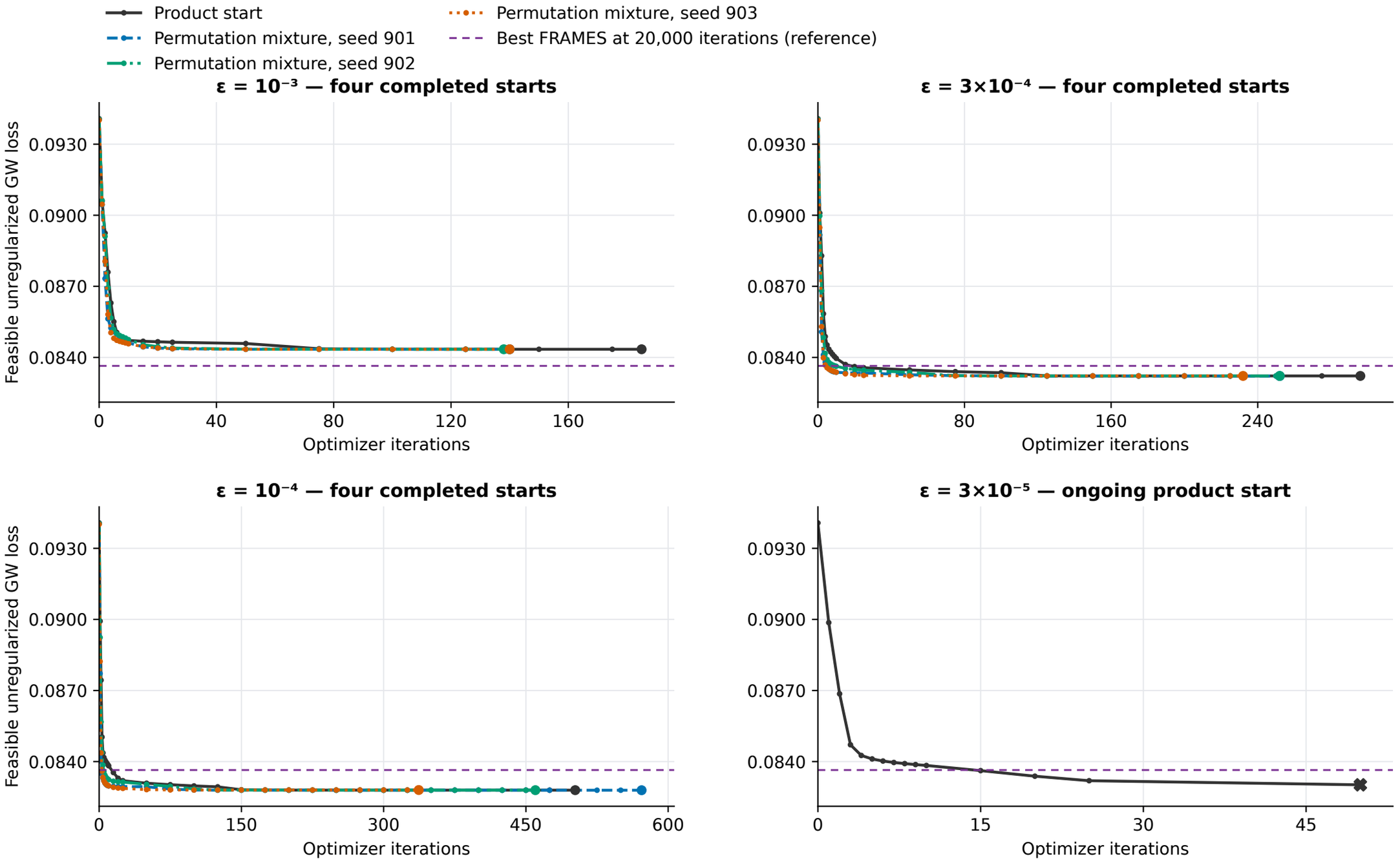


Dots are recorded diagnostics; × marks the last diagnostic of an unfinished run. All panels use the same loss scale.

Other starts at ε=3×10⁻⁵ and all ε=10⁻⁶ runs have no trajectories yet. Dashed purple lines are references, not POT runs.

SNARE-seq · 10,000 cells: every available POT trajectory

Full-gradient entropy-regularized GW updates · Sinkhorn tolerance 10^{-7} · 21 Oct, 09:58 Paris



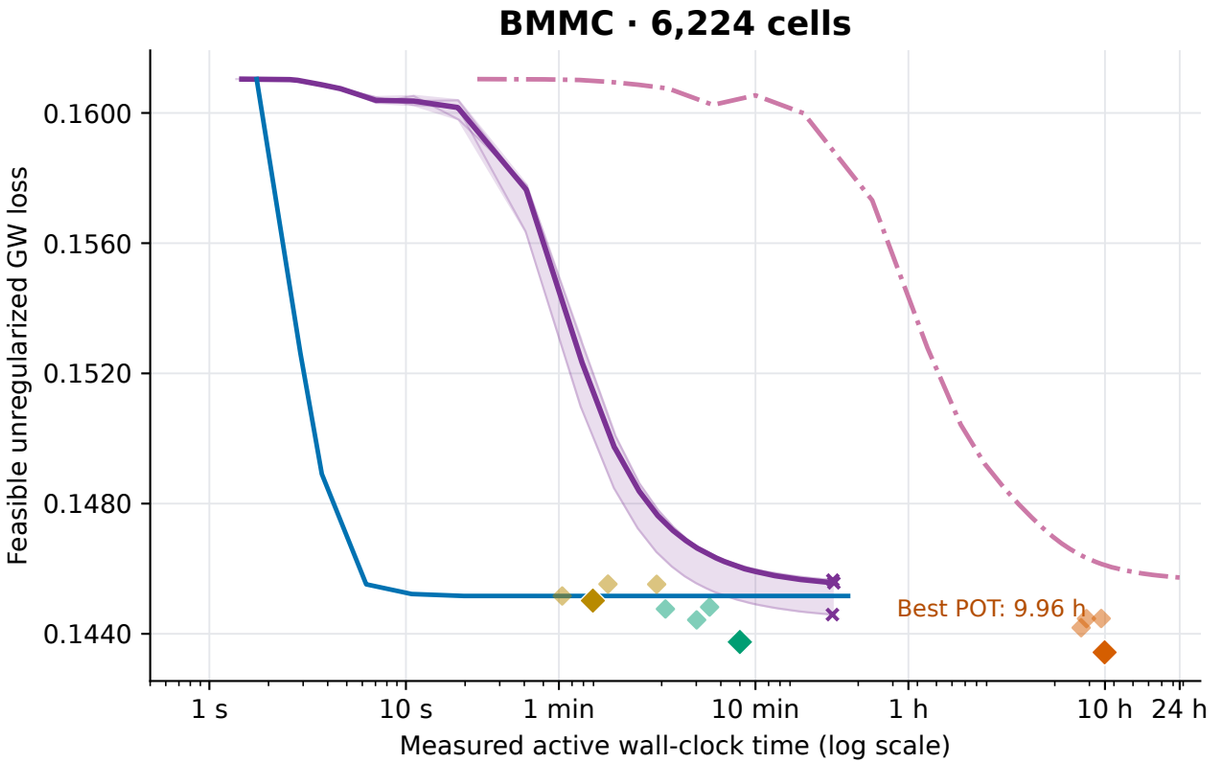
Dots are recorded diagnostics; × marks the last diagnostic of an unfinished run. All panels use the same loss scale.

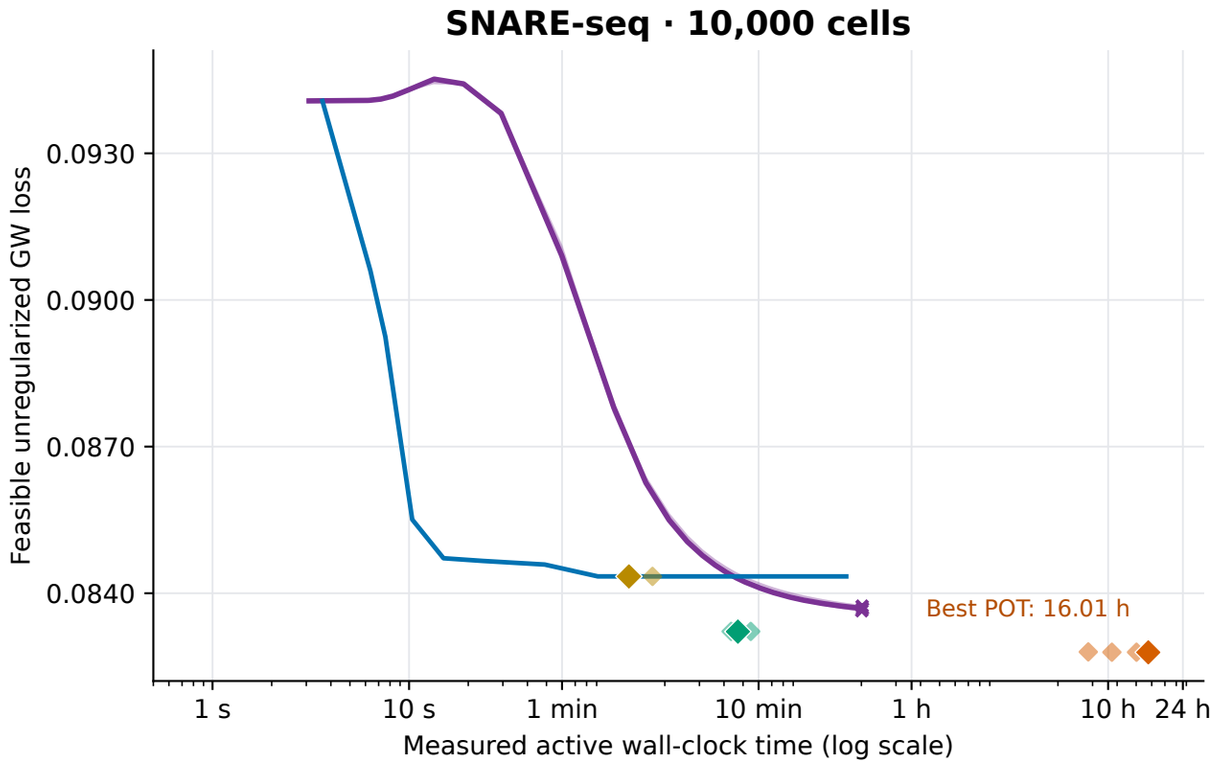
Other starts at ε=3×10⁻⁵ and all ε=10⁻⁶ runs have no trajectories yet. Dashed purple lines are references, not POT runs.

GW loss versus measured wall-clock time

FP64 · One A100 per GPU run; CPU run explicitly marked · Curves use recorded timestamps · Logarithmic time axis

FRAMES, 1,024 pairs, A100: median and range of 3 seeds
Full-gradient POT, ε=10⁻³, product start: 2,000 updates
FRAMES, 4,096 pairs, CPU: seed 811
POT sweep, ε=10⁻³: completed-run endpoints
POT sweep, ε=3×10⁻⁴: completed-run endpoints
POT sweep, ε=10⁻⁴: completed-run endpoints





Lines: timed replay (POT 2,000 updates; FRAMES 20,000). Diamonds: all completed starts in the entropy sweep; larger = lowest loss at that ε.

The sweep saved total runtime but no loss timestamps: diamonds are final results only, not reconstructed trajectories. Unfinished sweep runs omitted here.

Time includes data loading, optimization, loss diagnostics and checkpoint work; excludes queueing, prefactor tuning and runtime initialization.

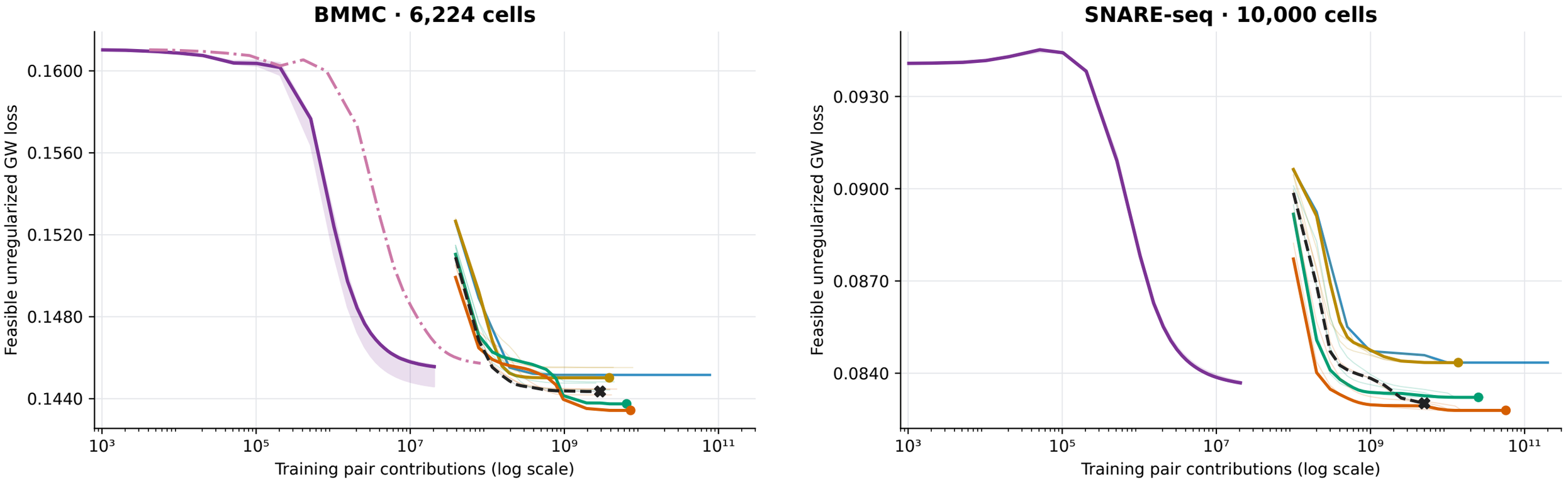
FRAMES time median/range interpolate only within shared observed time support. CPU and A100 timings are different hardware comparisons.

POT sweep starts vary; FRAMES starts from the product coupling. The 2,000-update POT replay has no early stopping; the sweep does.

GW loss versus training oracle work

Pair contributions at accepted updates · FRAMES: actual pair draws · Full-gradient POT: equivalent complete weighted sums

FRAMES, 1,024 pairs, A100: median and range of 3 seeds
Full-gradient POT, ε=10^{-3}, product start: 2,000 updates
FRAMES, 4,096 pairs, CPU: seed 811
POT sweep, ε=10^{-3}: best completed start
POT sweep, ε=3×10^{-4}: best completed start
POT sweep, ε=10^{-4}: best completed start
POT sweep, ε=3×10^{-5}: unfinished product start



After k accepted updates: FRAMES counts qk sampled pairs; full-gradient POT counts nmk weighted-pair equivalents (n × m coupling).

POT evaluates the full sum using matrix products; nm equivalents are not nm literal stochastic-oracle calls. This axis is not FLOPs or runtime.

Loss diagnostics and interrupted/retried updates are excluded. Sinkhorn iterations reuse the gradient and add no gradient calls.

Lines connect diagnostics; zero-work initialization is omitted. Faint POT lines: other starts. Unfinished endpoints: BMMC k=75; SNARE k=50.

Counting one batched gradient evaluation as one call instead gives k calls for either method, reproducing the iteration axis.